

On the Structure of the Least Favorable Prior Distributions

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What is This Talk About?

$$\max_{P_{\mathbf{X}} \in \mathcal{M}} I(\mathbf{X}; \mathbf{Y})$$

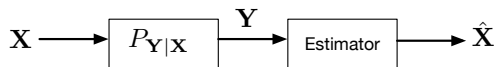
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We are interested in:

$$\max_{P_{\mathbf{X}} \in \mathcal{M}} \text{mmse}(\mathbf{X}|\mathbf{Y})$$

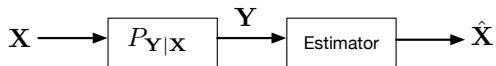
Definitions



where

- ▶ $\mathbf{X} \in \mathbb{R}^n$ and $\mathbf{Y} \in \mathbb{R}^k$;
- ▶ $(\mathbf{X}, \mathbf{Y}) \sim P_{\mathbf{XY}}$ with marginals $\mathbf{X} \sim P_{\mathbf{X}}$ and $\mathbf{Y} \sim P_{\mathbf{Y}}$;

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- ▶ Minimum Mean Square Error (MMSE)

$$\begin{aligned}\text{mmse}(\mathbf{X}|\mathbf{Y}) &\triangleq \inf_{f(\cdot): f-\text{measurable}} \mathbb{E} [\|\mathbf{X} - f(\mathbf{Y})\|^2] \\ &= \mathbb{E} [\|\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{Y}]\|^2] \\ &= \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}|\mathbf{X}}).\end{aligned}$$

Objective

- ▶ For a fixed $P_{Y|X}$

$$\sup_{P_X \in \mathcal{M}} \text{mmse}(P_X, P_{Y|X})$$

where $\mathcal{M}$ is a convex set of input distributions.

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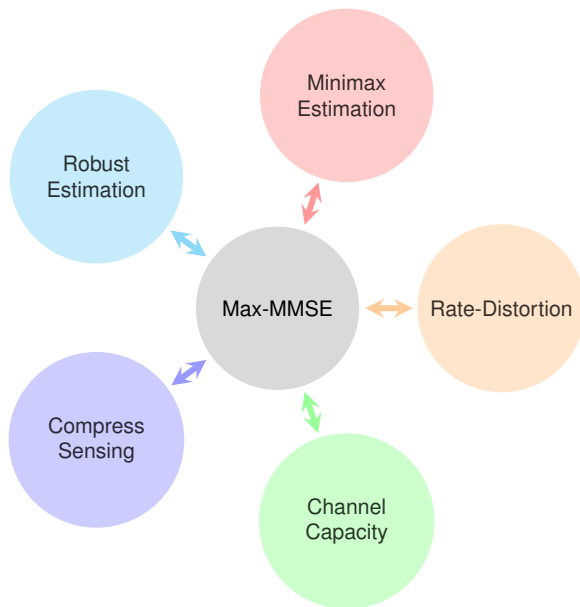
where $\mathcal{M}$ is a convex set of input distributions.

- ▶ A maximizer $P_{\mathbf{X}}^*$ is called a **least favorable prior distribution**.
- ▶ Examples of $\mathcal{M}$

$$\mathcal{M} = \{P_{\mathbf{X}} : \|\mathbf{X}\| \leq A\}$$

$$\mathcal{M} = \{P_{\mathbf{X}} : \mathbb{E}[f(\mathbf{X})] \leq \alpha\}$$

Why Is Max-MMSE an Interesting Problem?



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Examples:

- ▶ (Minimax Estimation¹)

$\mathcal{M} = \{P_{\mathbf{X}} : \mathbf{X} \in \mathcal{S} \text{ where } \mathcal{S} \text{ is a compact subset of } \mathbb{R}^n\}$

$$\max_{P_{\mathbf{X}} \in \mathcal{M}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}|\mathbf{X}}) \iff \inf_{f(\cdot)} \max_{\mathbf{x} \in \mathcal{S}} \mathbb{E} [\|\mathbf{x} - f(\mathbf{Y})\|^2] ;$$

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- ▶ (Capacity of Gaussian Channels via I-MMSE^{2,3})

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$$\begin{aligned} \max_{P_{\mathbf{X}} \in \mathcal{M}} I(\mathbf{X}; \sqrt{\text{snr}}\mathbf{X} + \mathbf{Z}) &= \max_{P_{\mathbf{X}} \in \mathcal{M}} \frac{1}{2} \int_0^{\text{snr}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}_\gamma|\mathbf{X}}) d\gamma \\ &\leq \frac{1}{2} \int_0^{\text{snr}} \max_{P_{\mathbf{X}} \in \mathcal{M}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}_\gamma|\mathbf{X}}) d\gamma. \end{aligned}$$

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- ▶ Robust Statistics (Fauß et al. 2018)
- ▶ Compress Sensing (Reeves and Donoho 2013)
- ▶ Arbitrary Channels via I-MMSE (Guo, Shamai (Shitz), and Verdú 2005)

Past Work

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 - ▶ Univariate case with a bounded support (i.e., $\mathcal{M} = \{P_X : X \in \mathcal{S} \subseteq \mathbb{R}^1 \text{ is compact}\}$)
 - ▶ If $x \mapsto R(x, P_X)$ is a real-analytic function, the P_X^* is discrete with finitely many point⁴;

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 - ▶ Sufficient and necessary conditions for optimality of two point distribution⁵.
 - ▶ Multivariate i.i.d. case is also known⁶.
- ▶ Questions:
 - ▶ What is the correct generalization to the multivariate case?
 - ▶ Are discrete input distributions still optimal in the multivariate case? If yes, how many points are there?
 - ▶ What about constraints of the form $\mathcal{M} = \{P_X : F(P_X) \leq c\}$ where $F(\cdot)$ is some operator?
 - ▶ Systematic approach
 - ▶ Is there a convergence algorithm?

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General Solution

Characterization of an Optimal Distribution

If $\mathcal{M} = \{P_{\mathbf{X}} : \mathbb{E}_{P_{\mathbf{X}}}[f(\mathbf{X})] \leq \alpha\}$. Then, $P_{\mathbf{X}}^*$ is optimal if and only if there exists $\lambda \geq 0$ and the following three conditions hold:

$$\lambda (\mathbb{E}_{P_{\mathbf{X}}^*}[f(\mathbf{X})] - \alpha) = 0, \quad (1)$$

$$R(\mathbf{x}, P_{\mathbf{X}}^*) - \lambda(f(\mathbf{x}) - \mathbb{E}_{P_{\mathbf{X}}^*}[f(\mathbf{X})]) \leq \text{mmse}(P_{\mathbf{X}}^*, P_{\mathbf{Y}|\mathbf{X}}), \quad \mathbf{x} \in \mathbb{R}^n, \quad (2)$$

$$R(\mathbf{x}, P_{\mathbf{X}}^*) - \lambda(f(\mathbf{x}) - \mathbb{E}_{P_{\mathbf{X}}^*}[f(\mathbf{X})]) = \text{mmse}(P_{\mathbf{X}}^*, P_{\mathbf{Y}|\mathbf{X}}), \quad \mathbf{x} \in \text{supp}(P_{\mathbf{X}}^*). \quad (3)$$

where

$$R(\mathbf{x}, P_{\mathbf{X}}) = \mathbb{E}[\|\mathbf{X} - \mathbb{E}_{P_{\mathbf{X}}}[\mathbf{X}|\mathbf{Y}]\|^2 \mid \mathbf{X} = \mathbf{x}].$$

When is the Support of an Optimal Input Distribution Bounded?

(Condition for An Unbounded Support)

If $f(\mathbf{x}) = o(\|\mathbf{x}\|^2)$, then the support of the optimizing distribution is *unbounded*.

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- ▶ note that this is independent of $P_{\mathbf{Y}|\mathbf{X}}$.

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- ▶ e.g., $f(\mathbf{x}) = \|\mathbf{x}\|^p$, $p \in (0, 2)$; and
- ▶ holds if $P_{\mathbf{Y}|\mathbf{X}}$ has a Gaussian tail.

The Support of an Optimal Input Distribution is 'Small'

(Structure of The Support)

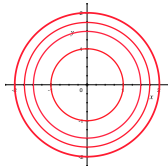
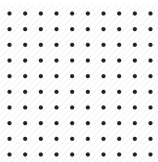
Suppose $f(\mathbf{x}) = o(\|\mathbf{x}\|^2)$ or $f(\mathbf{x}) = \omega(\|\mathbf{x}\|^2)$ and $R(\mathbf{x}, P_{\mathbf{x}}^*)$ and $f(\mathbf{x})$ are real-analytic. Then, the support of $P_{\mathbf{x}}^*$ is

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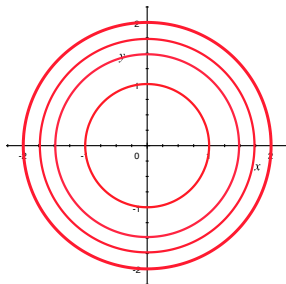
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- ▶ a nowhere dense set in $\mathbb{R}^n$;
- ▶ Lebesgue measure zero set;
- ▶ for $n = 1$, discrete; and
- ▶ $R(\mathbf{x}, P_{\mathbf{x}}^*)$ and $f(\mathbf{x})$ are orthogonally equivariant (i.e., only depend on $\|\mathbf{x}\|$) then the support of $P_{\mathbf{x}}^*$ is a countable union of concentric sphere $\bigcup_j \mathcal{C}(r_j)$ where $\mathcal{C}(r_j) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = r_j\}$



Algorithm

Observe the following:

$$\begin{aligned}\max_{P_{\mathbf{X}}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{X}|\mathbf{Y}}) &= \max_{P_{\mathbf{X}}} \mathbb{E} [\|\mathbf{X} - \mathbb{E}_{P_{\mathbf{X}}}[\mathbf{X}|\mathbf{Y}]\|^2] \\ &= \max_{P_{\mathbf{X}}} \inf_{f \text{ measurable}} \mathbb{E} [\|\mathbf{X} - f(\mathbf{Y})\|^2] \\ &= \max_{P_{\mathbf{X}}} \inf_{f(\mathbf{y}) = \mathbb{E}_{Q_{\mathbf{X}}}[\mathbf{X}|\mathbf{Y}=\mathbf{y}]} \mathbb{E} [\|\mathbf{X} - f(\mathbf{Y})\|^2] \\ &= \max_{P_{\mathbf{X}}} \inf_{Q_{\mathbf{X}}} \text{mmse}_{P_{\mathbf{X}}}(Q_{\mathbf{X}}, P_{\mathbf{Y}|\mathbf{X}})\end{aligned}$$

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Note that $P_{\mathbf{X}} \mapsto \text{mmse}_{P_{\mathbf{X}}}(Q_{\mathbf{X}}, P_{\mathbf{Y}|\mathbf{X}}) = \int R(\mathbf{x}, Q_{\mathbf{X}}) dP_{\mathbf{X}}(\mathbf{x})$ is linear in $P_{\mathbf{X}}$.

We can use this observation to design an algorithm. Specifically, what is called a Cutting Plane algorithm.

Example

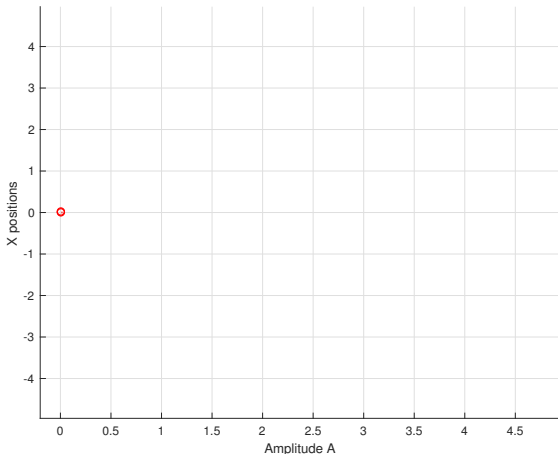
Consider

- ▶ $\max_{P_X: X \in [-A, A]} \text{mmse}(P_X, P_{Y|X})$
- ▶ where $P_{Y|X} = \mathcal{N}(x, 1)$;
- ▶ $I(X; \sqrt{\text{snr}}X + Z) = \frac{1}{2} \int_0^{\text{snr}} \text{mmse}(P_X, P_{Y_\gamma|X}) d\gamma$

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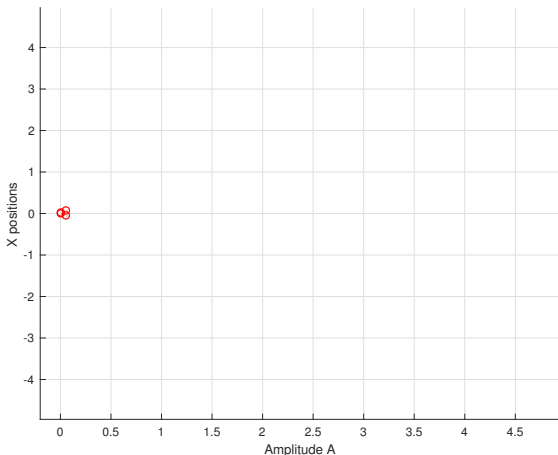
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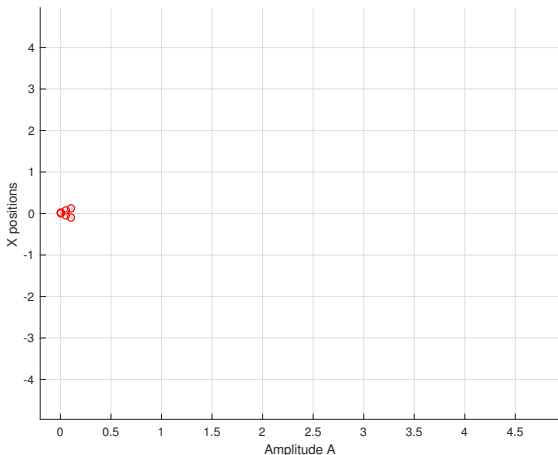
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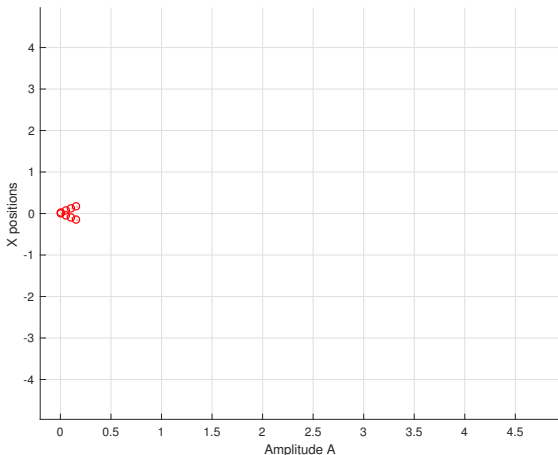
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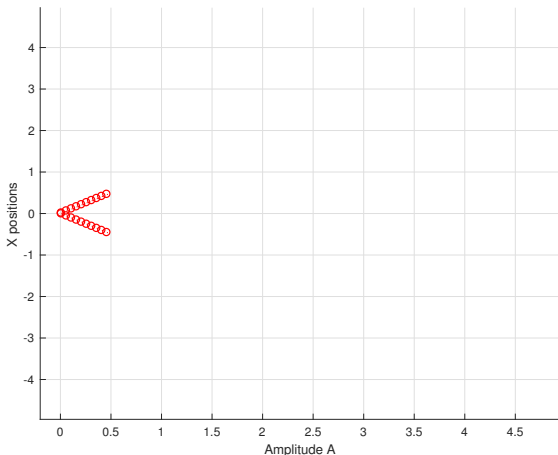
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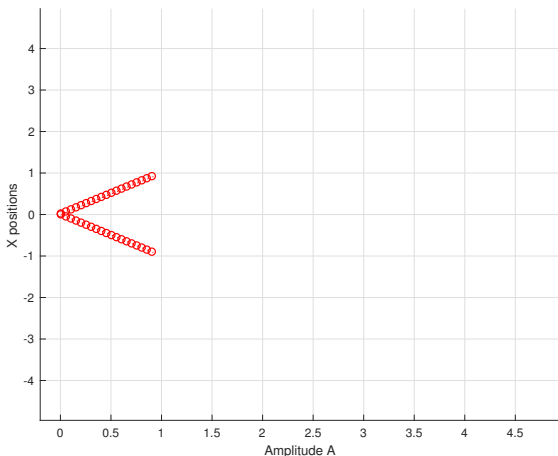
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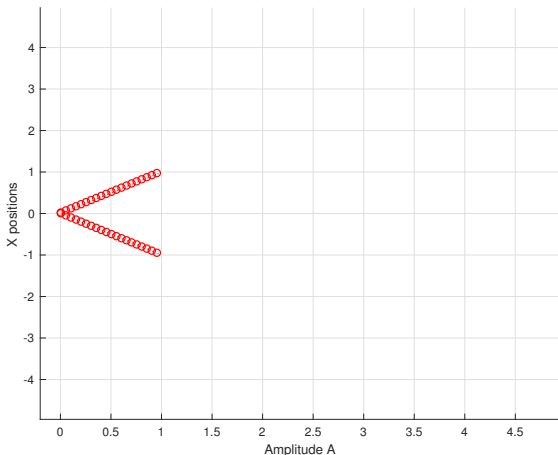
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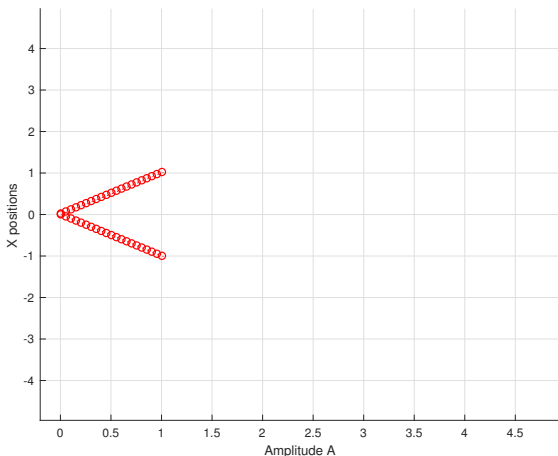
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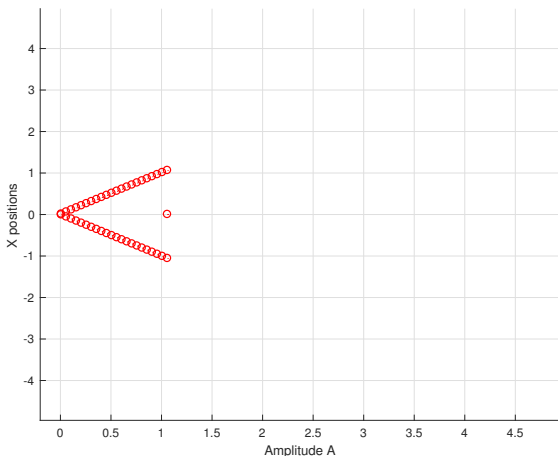
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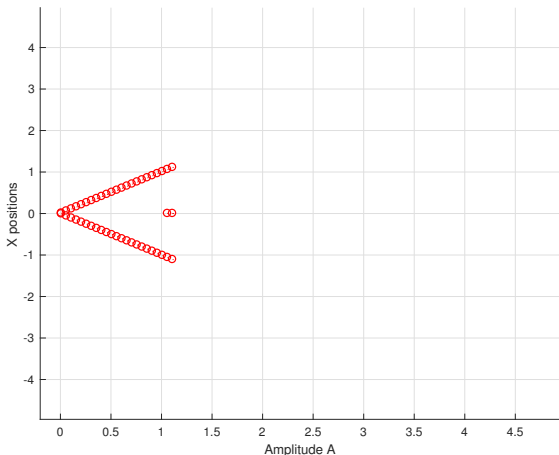
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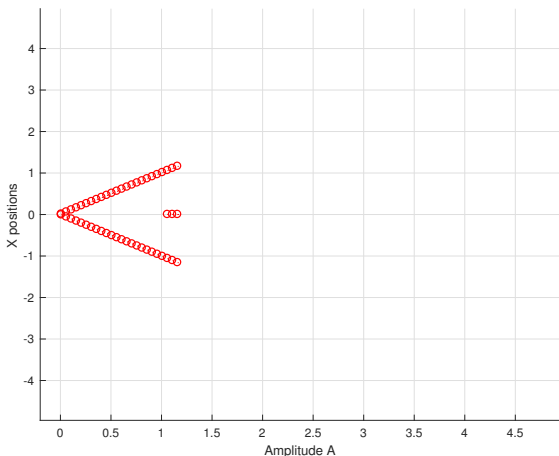
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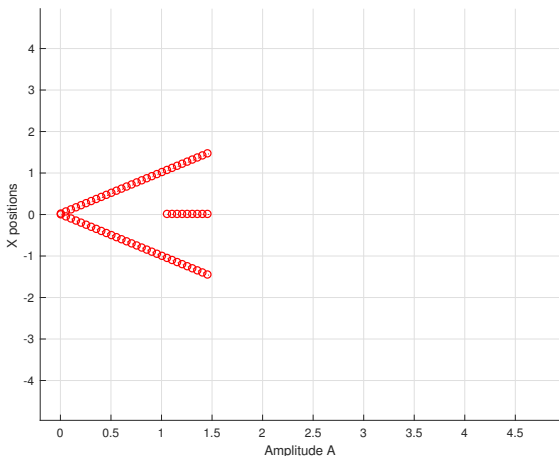
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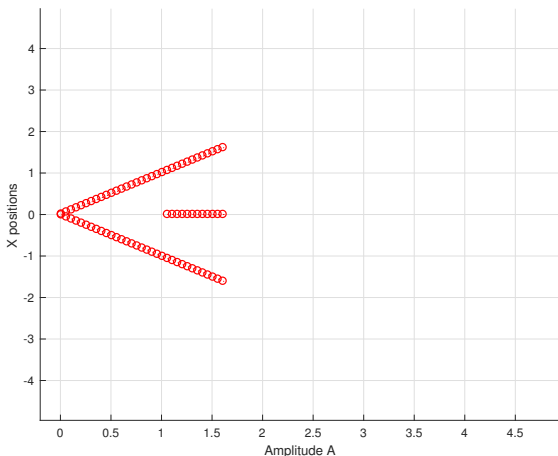
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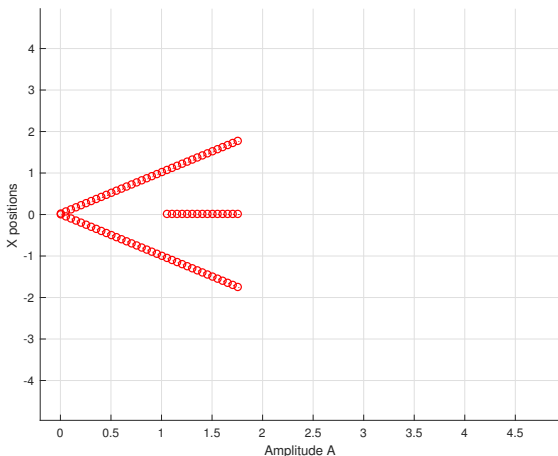
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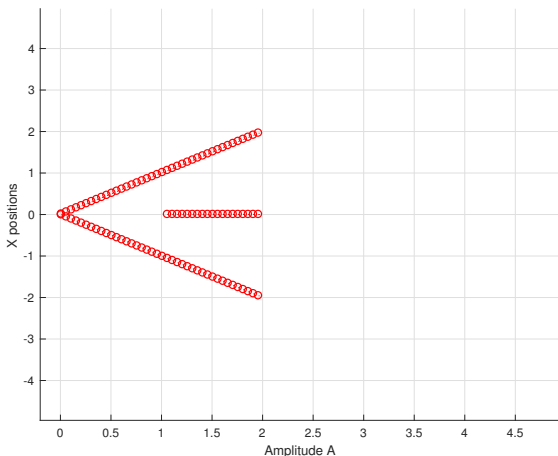
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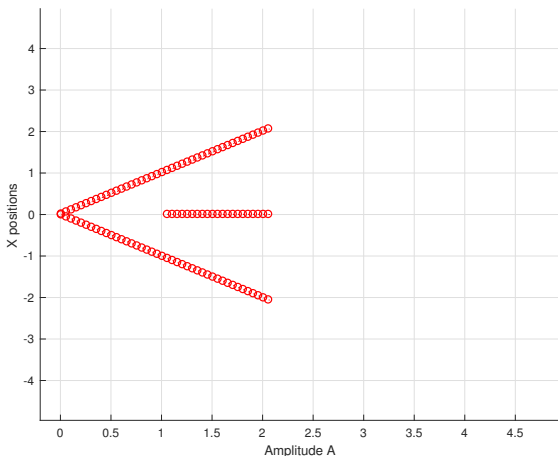
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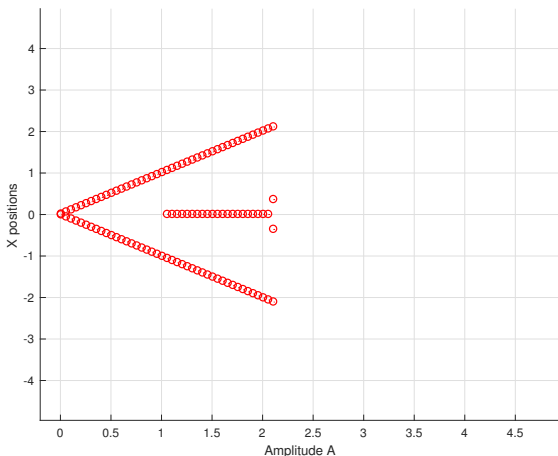
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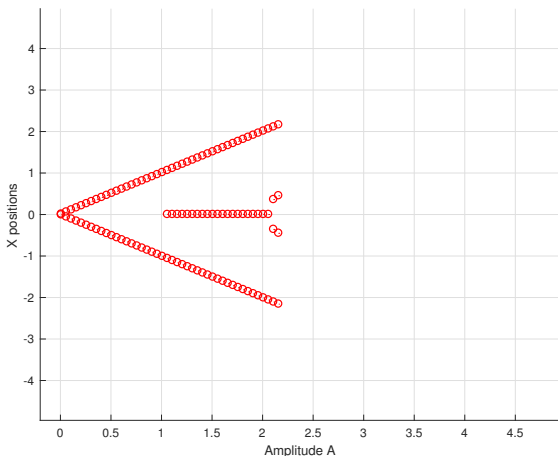
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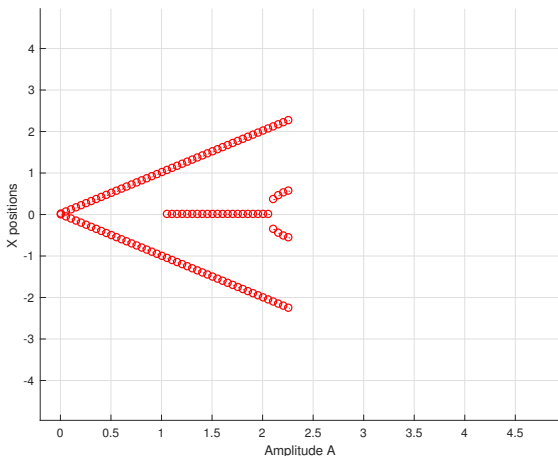
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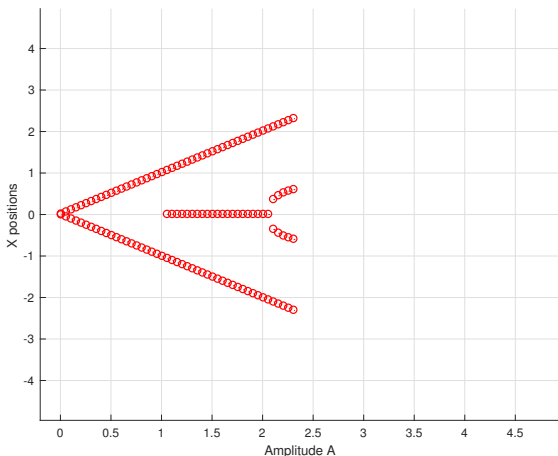
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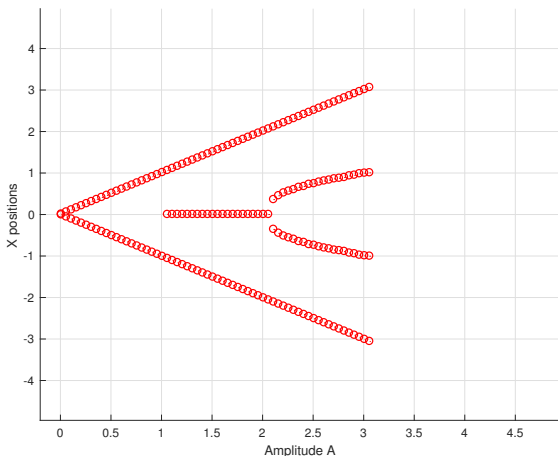
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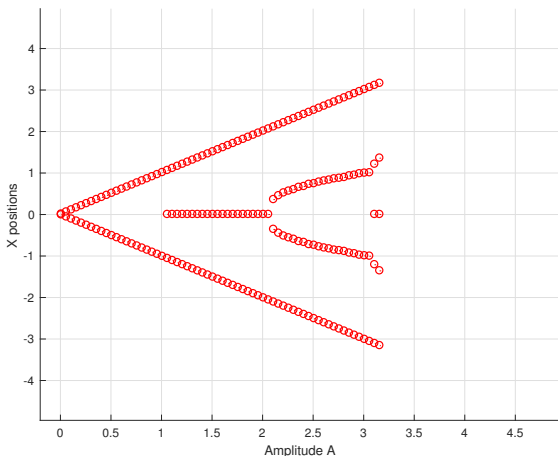
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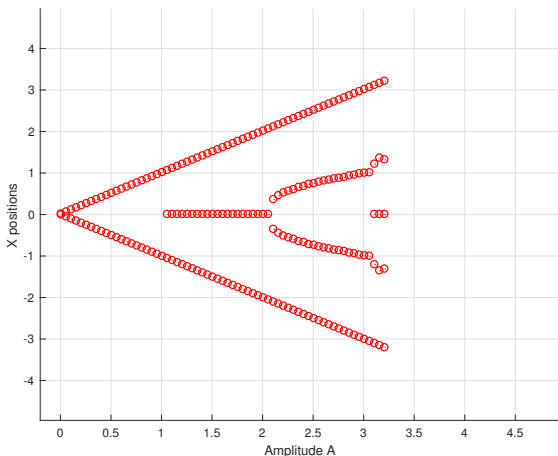
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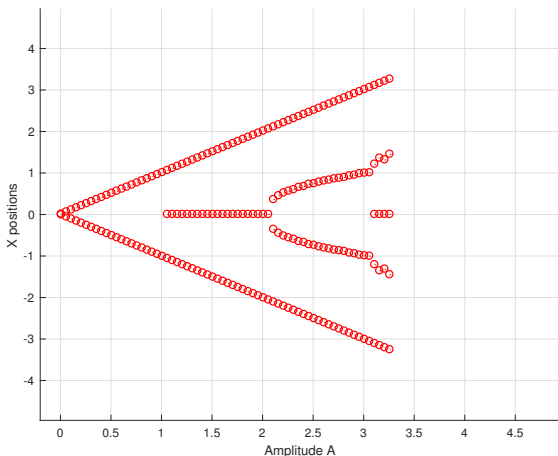
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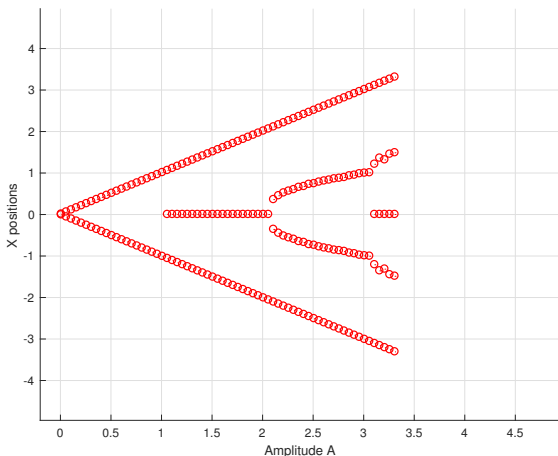
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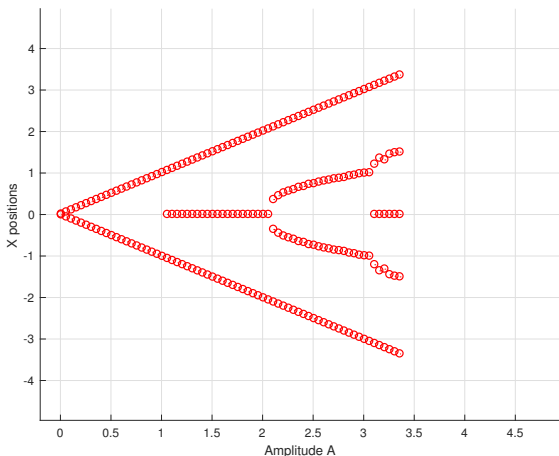
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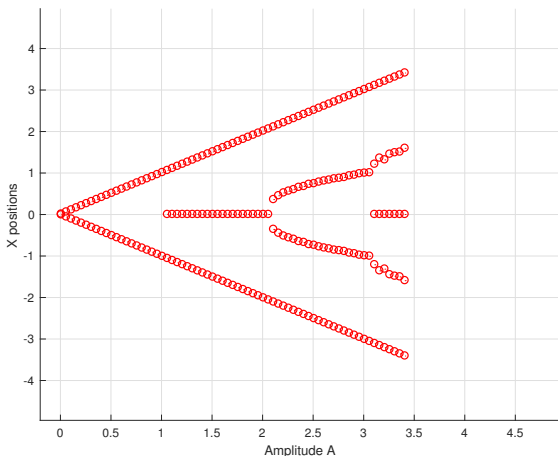
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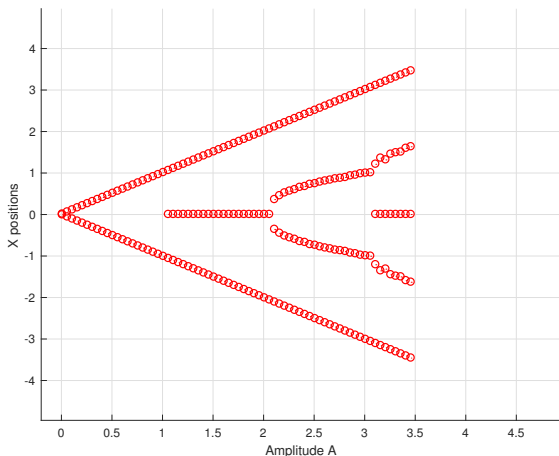
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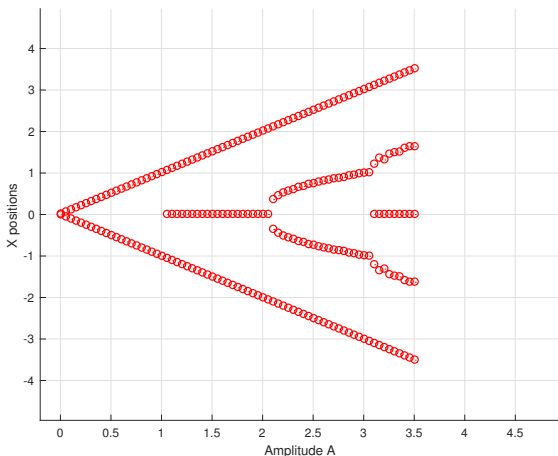
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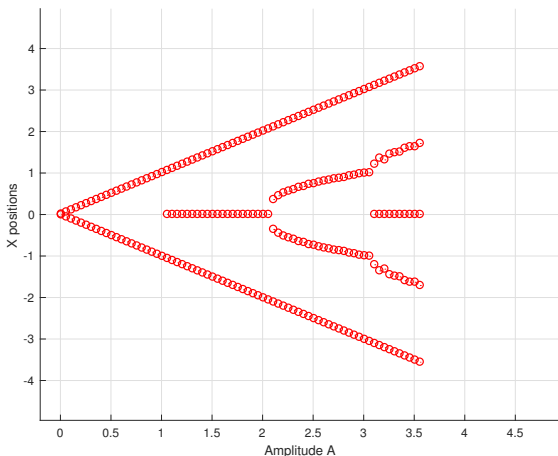
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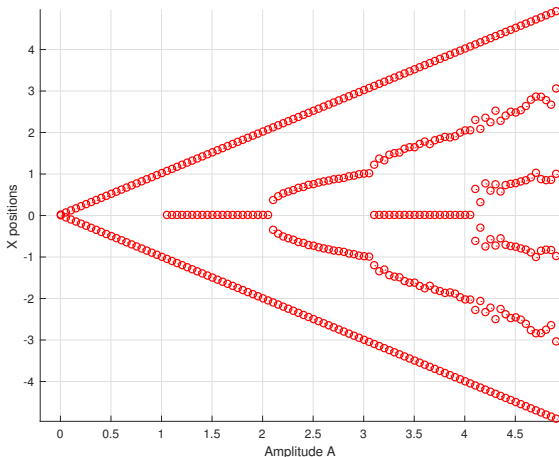
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Conclusion

- ▶ Integral equation that characterize an optimal $P_{\mathbf{x}}^{\star}$
- ▶ Conditions for bounded support
- ▶ Conditions when the support is small
- ▶ Algorithm based on cutting plane method

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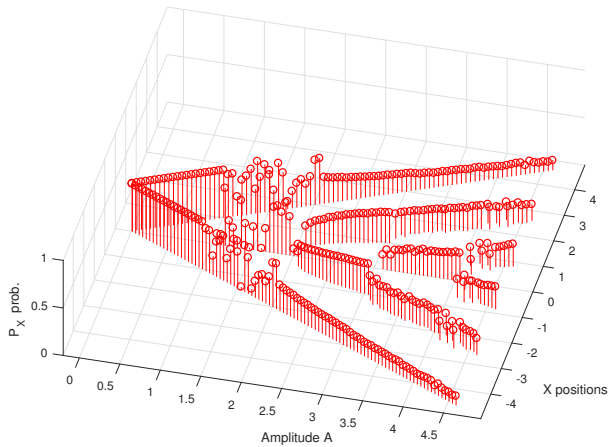
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Back Up Slides



Why Is Max-MMSE an Interesting Problem?

- ▶ Robust Statistics (Fauß et al. 2018)
- ▶ Compress Sensing (Reeves and Donoho 2013)
- ▶ Arbitrary Channels via I-MMSE (Guo, Shamai (Shitz), and Verdú 2005)

$$\begin{aligned}\max_{P_{\mathbf{X}} \in \mathcal{M}} I(\mathbf{X}; \sqrt{\text{snr}}\mathbf{X} + \mathbf{Z}) &= \max_{P_{\mathbf{X}} \in \mathcal{M}} \frac{1}{2} \int_0^{\text{snr}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}_\gamma|\mathbf{X}}) d\gamma \\ &\leq \frac{1}{2} \int_0^{\text{snr}} \max_{P_{\mathbf{X}} \in \mathcal{M}} \text{mmse}(P_{\mathbf{X}}, P_{\mathbf{Y}_\gamma|\mathbf{X}}) d\gamma.\end{aligned}$$

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- ▶ (Arbitrary Channels via I-MMSE⁷) For two real valued (X, U)

$$I(U; X) = \frac{1}{2} \int_0^\infty \text{mmse}(X|\sqrt{\gamma}X + Z) - \text{mmse}(X|\sqrt{\gamma}X + Z, U) d\gamma.$$

- ▶ Rate Distortion (Merhav 2011) Let $\mathbf{X}$ -be the source and $\hat{\mathbf{X}}$ a reconstruction for a distortion $d(\mathbf{x}, \hat{\mathbf{x}})$ and let $\Delta = d(\mathbf{X}, \hat{\mathbf{X}})$. Then, for all $s > 0$

$$R(D_s) = \inf_{P_{\hat{\mathbf{X}}}} \left(R_{P_{\hat{\mathbf{X}}}}(D_\infty) - \int_s^\infty \hat{s} \cdot \text{mmse}(\Delta|\mathbf{X}) \right).$$

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If f is complex analytic inside and on a regular curve $\mathcal{C}$ then

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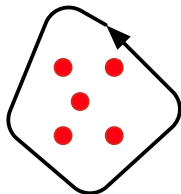
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